

Where $E_{res} = 80.3$ GeV is the target resonance energy, consistent with the W boson mass anomaly.

3. Time Domain (Pulse Duration)

To "calm" the vacuum and suppress quantum noise, the excitation pulse must have a precisely defined duration Δt . This duration is derived from the uncertainty principle and the stability condition.

Value:

$$\Delta t = 1.7 \times 10^{-15} \text{ s} \quad (1.7 \text{ fs})$$

Derivation: This time corresponds to the interval required to establish a coherent state within the digital lattice:

$$\Delta t \approx \frac{\hbar}{\Gamma_{width}} \cdot C_{stab}$$

Where Γ_{width} is the decay width of the bosons and C_{stab} is the stability correction factor.

4. Stability Threshold (Quantum Correction)

The bubble becomes stable only if the rate of vacuum fluctuations (noise) falls below a critical value δ_{vac} .

Value:

$$\delta_{vac} = 9.81 \times 10^{-5}$$

This dimensionless constant represents the synchronization tolerance. If the phase shift of the beams is more precise than this value, vacuum "locking" occurs.

5. Prediction: Inverse Asymmetry and Null Flux

Experimental confirmation of the theory relies on two key observations at the beam intersection point.

A. Asymmetry (Inverted Plot): Due to a 180° phase shift (π radians) in the digital structure, the predicted asymmetry A_{FB} at an energy of 80 GeV assumes the value:

$$A_{FB}(80 \text{ GeV}) = 0.05 \quad (\text{or inversion } -0.05)$$

This explains the anomaly measured in the CDF II and ATLAS experiments.

B. Null Photon Flux (Photon Depletion): Upon reaching the perfect angle $\theta = 28.18829^\circ$, complete destructive interference for electromagnetic radiation occurs. The photon flux Φ_γ drops to zero: